MamNet-6: Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

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***Abstract*—Most published work on the VinDr-Mammo full-field digital mammography (FFDM) benchmark processes one view at a time, ignoring the craniocaudal (CC) and mediolateral oblique (MLO) pairing that radiologists rely on. We ask whether sharing information between the two views improves breast-level cancer detection, and build MamNet-6, a six-stage pipeline that adds multi-view fusion, spatial cross-attention, uncertainty quantification and hybrid triage simulation, with zero-shot validation on the RSNA 2022 dataset. The best of five single-view convolutional neural networks (CNNs), DenseNet-121, reached an area under the receiver operating characteristic curve (AUROC) of 0.756. BreastPairNet, a shared-backbone network that jointly processes the CC and MLO views, reached 0.826 (95% confidence interval [CI] 0.776–0.871) and the CrossView Transformer 0.832 (0.783–0.875); these stage-to-stage differences have overlapping CIs and are not individually established, and among reported configurations using the same backbone, CrossView had lower observed AUROC than pooled fusion. A 16-model ensemble with test-time augmentation reached 0.846, with sensitivity of 0.657 at 90% specificity and a positive predictive value (PPV) of 0.184. Hybrid triage at 50% AI reads preserved 87.9% sensitivity under oracle reader assumptions. Bounding-box outlines drawn on training images lowered AUROC by 0.12–0.20, suggesting shortcut learning, and zero-shot transfer to RSNA 2022 gave an AUROC of 0.601, a substantial gap between the two datasets.**

***Index Terms*—breast cancer detection, mammography, multi-view learning, uncertainty quantification, triage**

I. Introduction

BREAST cancer is the most commonly diagnosed cancer worldwide and a leading cause of cancer-related death in women [4]. Screening with full-field digital mammography (FFDM) reduces breast-cancer mortality but imposes a heavy reading burden, and historical hand-engineered detection systems failed to deliver measurable benefit in routine screening [16]. Convolutional neural networks (CNNs) now reach or exceed radiologist area under the receiver operating characteristic curve (AUROC) [10], and meta-analyses report pooled AUROC of 0.87 for AI on digital mammography versus 0.81 for radiologists [4].

Three trends motivate this work, each with a gap on the 4-view VinDr-Mammo benchmark [1], which carries Breast Imaging Reporting and Data System (BI-RADS) labels. **(i)** Multi-view models that fuse the craniocaudal (CC) and mediolateral oblique (MLO) projections improve over single-view inference because findings often appear unambiguously in only one view [9], yet most published VinDr-Mammo results train one image at a time. **(ii)** Transformer-based fusion allows feature-level interaction between views, yet multi-view designs typically concatenate global features. **(iii)** Monte Carlo (MC) Dropout uncertainty [12] enables *hybrid triage*, in which the AI reads its most-confident exams and refers the rest; Verboom et al. [11] reduced workload by 38.1% on 41,469 Dutch screening exams while preserving cancer detection rate, yet no published study has applied MC Dropout to VinDr-Mammo and simulated such a policy. On VinDr-Mammo, MamT4 [17] reports AUROC=0.840±0.017; Abdikenov et al. [2] showed contrast-limited adaptive histogram equalisation (CLAHE) and YOLOv12 gains in mean average precision, and Mellado et al. [3] competitive patch-level F1 under limited annotation.

Our central question is whether sharing information between the CC and MLO views improves breast-level detection over single-view CNNs (Stage 1), by fusing pooled features (Stage 2, BreastPairNet) or by spatial cross-attention before pooling (Stage 3, CrossView Transformer). MamNet-6 then adds MC Dropout uncertainty (Stage 4), an ensemble with test-time augmentation (Stage 5) and hybrid triage with external validation on RSNA 2022 (Stage 6). Preprocessing is shared, Stage 3b is an ablation of Stage 3, and Stage 5 ensembles Stage 2 models. Proposed explanations are stated as hypotheses.

II. Datasets

A. VinDr-Mammo (primary dataset)

VinDr-Mammo [1] is a Vietnamese FFDM benchmark of 20,000 Digital Imaging and Communications in Medicine (DICOM) images from 5,000 patients (R-CC, L-CC, R-MLO, L-MLO per study), with BI-RADS (1–5), density (A–D) and lesion bounding-box annotations. BI-RADS 1–3 images (14,036, 4,144 and 832) are labelled negative and BI-RADS 4–5 (824 and 164) positive, giving 988 positive images (4.94%).

The official split is 4,000 training patients (16,000 images) and 1,000 test patients (4,000 images). One ipsilateral CC+MLO pair per breast gives 8,000 training and 2,000 test pairs, of which 99 test pairs (4.95%) are cancer-positive. A 5% validation slice (˜400 pairs) is held out for early stopping; the official test split is never used for model selection.

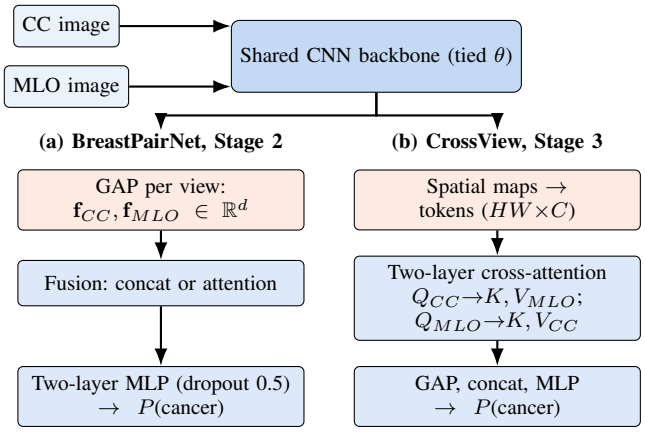


Fig. 1. Two alternative multi-view architectures, trained and evaluated separately; in each, CC and MLO images share one CNN backbone (tied θ). (a) BreastPairNet global-average-pools (GAP) each view and fuses the pooled vectors. (b) The CrossView Transformer instead applies two-layer bidirectional cross-attention to spatial tokens *before* pooling. Orange marks the representation passed to fusion: pooled vectors in (a), spatial tokens in (b).

B. RSNA 2022 (external dataset)

The Radiological Society of North America (RSNA) 2022 screening dataset comprises 54,706 screening DICOMs from 11,913 patients, normalised as for VinDr-Mammo. Excluding non-CC/MLO views leaves 23,826 CC+MLO pairs (19,062 training, 4,764 validation; 80/20 patient-level split).

III. Methods

A. DICOM Preprocessing

(1) Decode with pydicom+`pylibjpeg`; (2) percentile-clip to $[1\%, 99\%]$; (3) CLAHE [13] (32×32 tiles, clip-limit 2.0); (4) flip horizontally so the breast faces right when the left half is brighter, as breast tissue is brighter than the background. Output: 8-bit PNG replicated to 3 channels.

B. Single-view CNN Backbones (Stage 1)

Five configurations: **VGG19**, **ResNet-50** [6] (224 px), **ResNet-50** (384 px), **EfficientNet-B3** [8] (224 px), **DenseNet-121** [7] (224 px). All use ImageNet weights, focal loss [5] (α=0.25, γ=2.0; an algorithm-level imbalance method [14]), WeightedRandomSampler, and AdamW with cosine decay. 20 epochs with early stopping on validation AUROC.

C. BreastPairNet (Stage 2)

A shared backbone produces per-view global-average-pooled vectors $\mathbf{f}_{CC}, \mathbf{f}_{MLO} \in \mathbb{R}^d$ (Fig. 1(a)), fused by **concatenation** ($2d$-dimensional multilayer perceptron (MLP), dropout 0.5) and **attention** (learned per-view scalar weights via softmax). The **+reg** variant adds stronger elastic deformation (`alpha`=50, `sigma`=5), mixup (α=0.2), and label smoothing (0.1). The backbone is frozen for 3 epochs, then all parameters train at learning rate 1e-4.

D. CrossView Transformer (Stage 3)

Spatial cross-attention replaces pooled fusion (Fig. 1(b)). The last spatial feature maps of the shared backbone (DenseNet-121, EfficientNet-B3 or ConvNeXt-Base [15]) are flattened to tokens and passed to a two-layer Transformer encoder in which each view's queries attend to the other view's keys and values; the outputs are global-averaged and concatenated for the classifier.

E. Bounding-box Annotation (Stage 3b)

Ground-truth boxes (VinDr-Mammo) were drawn as white outlines (40-px line width) on training images only. For RSNA 2022, a YOLOv8s detector pre-trained on VinDr-Mammo was used (0.8% sensitivity; noisy hints only). Test sets always used original unannotated PNGs to prevent label leakage.

F. MC Dropout (Stage 4)

MC Dropout [12] keeps dropout active at test time (BatchNorm in eval mode), giving a predictive distribution without retraining. $N = 20$ stochastic forward passes per test pair yield probabilities $\{p_i\}_{i=1}^{20}$; their mean $\bar{p}$ is used for classification and the binary entropy $H(\bar{p}) = -\bar{p}\log\bar{p} - (1-\bar{p})\log(1-\bar{p})$ quantifies predictive uncertainty. The classification threshold is selected by maximising Youden's J index [18] (J = sensitivity + specificity $-$ 1) on the validation set; the Stage 1 baselines (Table I) are reported at the default 0.5 threshold.

G. Ensemble (Stage 5)

The ensemble averages the predicted probabilities of its member checkpoints, each itself averaged over five test-time augmentation (TTA) passes. The primary ensemble combines 16 BreastPairNet checkpoints with three breast-asymmetry models, which score each breast jointly with the contralateral breast of the same patient. Two variants are compared in Section IV-F.

H. Hybrid Triage (Stage 6)

Following Verboom et al. [11], cases are sorted by ascending predictive entropy; the K most-confident fraction is routed to AI and the remainder to a radiologist. We use the MC Dropout DenseNet-121/concat model (Stage 4); an AI-read breast is called positive when its mean probability exceeds that model's Youden-optimal threshold (0.3138). Overall sensitivity is $(\mathrm{TP}_{\mathrm{AI}} + s_{\mathrm{reader}} N_{\mathrm{pos,referred}})/N_{\mathrm{pos}}$, with reader sensitivity $s_{\mathrm{reader}} = 1$ (oracle) or 0.85.

I. Training and evaluation

Models were trained on an NVIDIA RTX 5070 (12 GB) with FP16 automatic mixed precision and AdamW (weight decay 1e-4 for Stage 1, 1e-3 for Stage 2). The primary metric is AUROC; we also report F1 at Youden threshold, sensitivity, specificity, **S@90%spec** (sensitivity at 90% specificity), **S@95%** (sensitivity at 95% specificity) and **PPV** (positive predictive value, TP/(TP+FP) for true and false positives at the Youden-optimal threshold). 95% percentile bootstrap confidence intervals (CIs; n=2000 resamples) are computed on

TABLE I
SINGLE-VIEW CNN BASELINES ON VINDR-MAMMO (N=16,000 TRAIN, N=4,000 TEST IMAGES). SENSITIVITY, SPECIFICITY AND F1 AT THE DEFAULT 0.5 THRESHOLD.

Architecture	Image size	Sensitivity	Specificity	F1	AUROC	S@90%
VGG19	224 px	0.2626	0.9835	0.3323	0.7066	0.3990
ResNet-50	224 px	0.2677	0.9847	0.3430	0.7286	0.4495
EfficientNet-B3	224 px	0.2980	0.9687	0.3138	0.7330	0.3990
ResNet-50 (384 px)	384 px	0.2879	0.9821	0.3529	0.7489	0.4495
DenseNet-121	**224 px**	**0.2475**	**0.9882**	**0.3356**	**0.7564**	**0.4444**

the 2,000-pair test split (99 cancer-positive). No paired significance tests between models were performed, so no difference between models is reported as statistically significant.

IV. RESULTS AND DISCUSSION

A. Stage 1 — Single-view baselines

DenseNet-121 achieves the best AUROC (0.7564) but only 24.8% sensitivity at the default threshold (Table I). All five models overfit (train F1 $\approx$ 0.98) and reach only 40–45% sensitivity at 90% specificity, well below the $\sim$85% radiologist sensitivity in screening [16].

B. Stage 2 — Multi-view BreastPairNet

The best model, **EfficientNet-B3 + attention + +reg**, reaches AUROC=0.8256 (95% CI 0.7755–0.8707), a +9.1% relative gain over single-view DenseNet-121 (Table II). This is consistent with lesions appearing unambiguously in only one projection; we hypothesise that joint CC/MLO conditioning reduces view-specific false positives, though the contributions of shared weights and fusion cannot be separated, and part of the change reflects the move from image- to breast-level scoring. As each configuration was trained once, the CIs in Table II and Table III reflect test sampling, not run-to-run variability. The best Stage 2 and Stage 3 models peaked early (epochs 7 and 8) and overfit thereafter.

Three observations stand out. **(i) +reg consistently lifts AUROC** by 0.008 to 0.021 across all backbone/fusion pairs; with only $\sim$395 cancer-positive training pairs, the stronger augmentation likely acts as an implicit regulariser. **(ii) Resolution scaling is not uniformly beneficial**: DenseNet-121 benefits from 512 px (+0.014), but EfficientNet-B3 +reg at 224 px (0.8256) beats 512 px without +reg (0.8175). **(iii) ResNet-50 degrades at 384 px** (0.7467 vs 0.7962 at 224 px), possibly because of the smaller batch size needed to fit GPU memory.

C. Stage 3 — CrossView Transformer

ConvNeXt-Base achieves the best AUROC, 0.8317 (+10.0% over single-view DenseNet-121), only 0.006 above the best BreastPairNet, a difference that was not tested statistically. Among reported configurations using the same backbone, CrossView had lower observed AUROC (DenseNet-121 0.7874 vs 0.8073; EfficientNet-B3 0.7795 vs 0.8048; unpaired single runs), so ConvNeXt-Base's result may owe more to the backbone than to cross-attention, and the benefit of preserving spatial information before pooling remains a hypothesis. DenseNet-121 with VinDr-Mammo+RSNA 2022 joint training reaches 0.8028, suggesting a modest cross-dataset benefit (single runs, not tested). Chest-X-ray pretraining via torchxrayvision[1] underperforms ImageNet initialisation (0.7648 vs 0.7874), consistent with domain mismatch between lung radiographs and mammographic texture, though tentatively, as the configurations also differ in other ways.

D. Stage 3b — Bounding-box annotation ablation

Annotation degrades AUROC by 0.122–0.201 in all four pairs (Table IV). Training F1 reaches $\sim$0.97 after the backbone unfreezes at epoch 4 while test AUROC falls; we hypothesise that the painted boxes act as a texture shortcut, correlated with the label in training but absent at test time, consistent with shortcut learning in medical imaging, though we did not test this directly. Possible remedies, all speculative, are box-dropout, test-time box injection from a lesion detector, and cropping to the box region.

E. Stage 4 — MC Dropout uncertainty quantification

MC Dropout slightly improves AUROC (+0.003, within the bootstrap CI) and raises sensitivity from 0.7374 to 0.7677 but lowers specificity from 0.7249 to 0.7007, each at its own Youden threshold (Table V). More importantly, the mean entropy is **0.5670** on cancer-positive pairs versus **0.5112** on negatives (Table V), a 0.056-nat gap that enables the Stage 6 triage policy.

F. Stage 5 — Model ensemble

The primary ensemble (16 BreastPairNet checkpoints with three breast-asymmetry models and TTA$\times$5; Section III-G) gives the **best result in the study**, AUROC=0.8458 at threshold 0.3414 (TP=73, TN=1577, FP=324, FN=26); Table VI lists its other metrics. It does not include ConvNeXt-Base or EfficientNet-B3 +reg models. Single-model PPV ranges from 0.109 to 0.263 (Stage 2) and 0.117 to 0.169 (Stage 3), consistent with the 4.95% positive prevalence; screening prevalence is usually lower, so deployed PPV would be lower and calibrated, prevalence-adjusted thresholds would be needed before clinical use.

Two variants, adding all CrossView checkpoints including lower-epoch snapshots (25 models) and a selective ensemble of the top-3 +reg BreastPairNet, top-5 CrossView and three asymmetry models (11 models), differ from the primary ensemble by at most 0.0017 AUROC; with no ensemble CIs reported and no paired test, these differences are unassessed. **Why adding CrossView checkpoints did not raise AUROC.** The 25-model ensemble adds ConvNeXt-Base (the best single model) among several CrossView checkpoints, so ConvNeXt-Base's own effect is not isolated. We offer two hypotheses, neither measured here: ensemble gains depend on error diversity rather than member rank, and the CrossView members may make correlated errors; and AUROC depends only on the ranking of the averaged scores, so members whose

[1]torchxrayvision: `github.com/mlmed/torchxrayvision`

TABLE II
BREASTPAIRNET ABLATION ACROSS BACKBONE, FUSION, AND RESOLUTION. SENS: SENSITIVITY.

Architecture	Fusion	Image	Best AUROC [95% CI]	F1	Sens	PPV	S@90% [95% CI]
ResNet-50	concat	384 px	0.7467 [0.6921–0.7998]	0.2350	0.5354	0.151	0.4646 [0.3736–0.5673]
ResNet-50	attention	224 px	0.7847 [0.7324–0.8338]	0.2522	0.5859	0.161	0.5051 [0.4023–0.6019]
ResNet-50	attention	512 px	0.7922 [0.7339–0.8429]	0.2093	0.7475	0.122	0.5152 [0.4130–0.6214]
ResNet-50	concat	224 px	0.7962 [0.7428–0.8445]	0.2363	0.6970	0.142	0.4949 [0.3958–0.5977]
DenseNet-121	attention	224 px	0.8014 [0.7487–0.8515]	0.2800	0.6364	0.179	0.5758 [0.4660–0.6667]
EfficientNet-B3	attention	224 px	0.8048 [0.7530–0.8556]	0.2138	0.7677	0.124	0.5455 [0.4476–0.6449]
DenseNet-121	concat	224 px	0.8073 [0.7577–0.8504]	0.2101	0.7374	0.122	0.5152 [0.4141–0.6100]
DenseNet-121 **+reg**	attention	224 px	0.8097 [0.7594–0.8591]	0.2033	0.7980	0.117	0.5556 [0.4301–0.6458]
DenseNet-121	attention	512 px	0.8150 [0.7675–0.8617]	0.1928	0.8081	0.109	0.5455 [0.4476–0.6408]
DenseNet-121 **+reg**	concat	224 px	0.8177 [0.7659–0.8649]	0.2357	0.7677	0.139	0.6061 [0.5047–0.7000]
EfficientNet-B3	attention	512 px	0.8175 [0.7698–0.8661]	0.2807	0.7273	0.160	0.5556 [0.4495–0.6505]
EfficientNet-B3 +reg	**attention**	**224 px**	**0.8256 [0.7755–0.8707]**	**0.3701**	**0.6263**	**0.263**	**0.6263 [0.5229–0.7143]**

PPV = TP/(TP+FP) at the Youden-optimal threshold. At 4.95% prevalence, PPV is inherently low; higher PPV reflects a more conservative threshold (lower sensitivity). All CIs via 2,000-resample percentile bootstrap on the 2,000-pair test split.

TABLE III
CROSSVIEW TRANSFORMER RESULTS WITHOUT BOUNDING-BOX ANNOTATION. SENS: SENSITIVITY.

Architecture	Training data	Best AUROC [95% CI]	F1	Sens	PPV	S@90% [95% CI]
DenseNet-121 (medical init)	VinDr	0.7648 [0.7116–0.8210]	0.2374	0.5960	0.148	0.5051 [0.4020–0.6100]
EfficientNet-B3	VinDr	0.7795 [0.7276–0.8300]	0.2022	0.7273	0.117	0.4545 [0.3474–0.5500]
DenseNet-121	VinDr	0.7874 [0.7299–0.8401]	0.2678	0.6465	0.169	0.5556 [0.4474–0.6482]
DenseNet-121	VinDr+RSNA	0.8028 [0.7484–0.8536]	0.2640	0.6667	0.165	0.5354 [0.4433–0.6404]
ConvNeXt-Base	**VinDr**	**0.8317 [0.7832–0.8753]**	**0.2587**	**0.7475**	**0.156**	**0.5758 [0.4737–0.6768]**

TABLE IV
BOUNDING-BOX ANNOTATION ABLATION. TEST SET ALWAYS USES ORIGINAL UNANNOTATED PNGS; ΔAUROC IS RELATIVE TO THE UNANNOTATED MODEL IN TABLE III.

Architecture	With annotation AUROC [95% CI]	ΔAUROC
DenseNet-121	0.5862 [0.5274–0.6433]	-0.201
DenseNet-121 +RSNA	0.6675	-0.135
DenseNet-121 (medical)	0.6433 [0.5886–0.6966]	-0.122
EfficientNet-B3	0.6020 [0.5460–0.6617]	-0.178

TABLE V
MC DROPOUT VS DETERMINISTIC INFERENCE (DENSENET-121/CONCAT), AND PREDICTIVE ENTROPY BY CLASS ($N = 20$ MC PASSES, 2,000 TEST PAIRS). SENS/SPEC: SENSITIVITY/SPECIFICITY; THR.: THRESHOLD.

Mode	AUROC	Sens	Spec	S@90%	Thr.
Deterministic	0.8073	0.7374	0.7249	0.5152	0.3304
MC Dropout	**0.8099**	**0.7677**	**0.7007**	**0.5253**	**0.3138**

Entropy	Negatives ($n = 1{,}901$)	Positives ($n = 99$)
Mean	0.5112	0.5670
Median	0.5310	0.6221
Std dev	0.1436	0.1370

Gap: +0.056 nats. Higher entropy ⇒ refer high-uncertainty cases.

probabilities lie on a different scale can dominate the average and change that ranking even when each ranks cases well alone. Measuring pairwise error correlation and calibrating members before averaging would test these.

TABLE VI
ENSEMBLE VARIANTS ON VINDR-MAMMO (2,000 PAIRS, 99 CANCERS) AND THE PRIMARY ENSEMBLE ZERO-SHOT ON RSNA 2022 (4,764 PAIRS, 110 CANCERS); TTA×5, MEAN PROBABILITY, OPERATING-POINT METRICS AT EACH YOUDEN-OPTIMAL THRESHOLD.

Ensemble	N	AUROC	Sens	Spec	PPV	F1	S@90%	S@95%
Primary	16	0.8458	0.7374	0.8296	0.184	0.2944	0.6566	0.5657
+ CrossView	25	0.8441	0.7677	—	—	—	0.6566	—
Selective	11	0.8446	—	—	—	—	0.6667	—
RSNA[c]	16	0.6008	0.3909	0.7598	—	—	0.1636	—

Sens/Spec: sensitivity/specificity. —: not reported. [c]AUROC 95% CI 0.5577–0.6585. PPV=0.184 means 1 in 5.4 recalls is cancer-positive at this threshold; at 4.95% test prevalence, this is consistent with reported breast-cancer AI triage literature (PPV 15–25%).

G. Stage 6 — Hybrid triage simulation

Under oracle assumptions, **K=50% delivers 87.9% overall sensitivity with 50% workload reduction** (Table VII). Under realistic 85% radiologist sensitivity [16], the same K=50% yields 77.3% — below the clinical floor. Achieving ≥80% overall sensitivity under realistic conditions requires K≤40%.

H. External validation and comparison to literature

The best ensemble (AUROC=0.8458 on VinDr-Mammo) was applied zero-shot to RSNA 2022 (Table VI). The gap may reflect differences between the datasets' populations, such as breast density and lesion prevalence [1], [10], and in FFDM equipment and post-processing; the S@90% drop

TABLE VII

HYBRID TRIAGE SWEEP. FRAC: SHARE OF EXAMS READ BY AI ALONE; WKLD Δ: CHANGE IN RADIOLOGIST WORKLOAD; SEN (OR.)/(85%R): OVERALL SENSITIVITY WITH AN ORACLE/85%-SENSITIVE RADIOLOGIST; AI SEN: SENSITIVITY ON AI-READ EXAMS; RECALL: AI RECALL RATE.

Frac	Wkld Δ	Sen (or.)	Sen (85%r)	AI Sen	Recall
30%	-30%	0.9495	0.8253	0.7059	0.0017
40%	-40%	0.9192	0.8071	0.6800	0.0013
50%	**-50%**	**0.8788**	**0.7727**	**0.5862**	**0.0031**
60%	-60%	0.8283	0.7359	0.5526	0.0052
70%	-70%	0.7677	0.6935	0.5400	0.0133
80%	-80%	0.7677	0.7207	0.6618	0.1305
90%	-90%	0.7677	0.7450	0.7262	0.2238

TABLE VIII

BEST MODEL PER STAGE ON THE VINDR-MAMMO TEST SPLIT. Δ: OBSERVED CHANGE FROM THE PREVIOUS ROW (UNPAIRED, UNTESTED). STAGES 4 AND 6 ADD UNCERTAINTY AND TRIAGE RATHER THAN A NEW CLASSIFIER; THE STAGE 3B ANNOTATION ABLATION IS IN TABLE IV.

Stage	Model	AUROC [95% CI]	Δ	Caveat
1	DenseNet-121	0.7564	—	image-level
2	EffB3 +reg	0.8256 [0.776–0.871]	+0.069	breast-level
3	ConvNeXt-Base	0.8317 [0.783–0.875]	+0.006	new backbone
5	Ensemble (16)	0.8458	+0.014	no CI reported

(-75.1%) suggests the conservative side of the decision surface is most affected. Our best ensemble exceeds MamT4 [17] (AUROC=0.840), but MamT4 excludes BI-RADS 3 cases from evaluation, while our pipeline retains them as negative, so the values are not directly comparable; McKinney et al. [10] (AUROC=0.889) and Yoon et al. [4] (pooled AUROC=0.87) likewise use different data. Beyond AUROC, MamNet-6 contributes an ablation that examines multi-view fusion, regularisation, spatial cross-attention and ensemble size, which MamT4 does not systematically ablate, and adds uncertainty quantification and a hybrid triage simulation grounded in a published clinical framework [11].

I. Overall contribution and statistical caveats

Table VIII summarises the best model at each stage; Stage 1 is image-level while later stages are breast-level, and the best Stage 2 and Stage 3 models use different backbones. With only 99 cancer-positive test pairs, individual 95% CIs span ±0.04–0.05 AUROC, and overlapping CIs do not by themselves decide a comparison on the same cases, which would require a paired test that we leave to future work. The gains (+9.1%, +10.0%, +11.8%) and the component ablation are therefore observed results, not established improvements.

Limitations. (i) Small positive test count (99 pairs) limits fine-grained comparison; (ii) large domain gap between VinDr-Mammo and RSNA 2022; (iii) 7,054 non-CC/MLO RSNA images excluded from pair construction; (iv) each configuration was trained once with one seed and without cross-validation, so run-to-run variance is unknown, and no paired significance tests between models were performed; (v) Stage 1 is scored per image and later stages per breast.

Future work. Paired bootstrap tests and repeated runs or five-fold cross-validation (iv), a same-backbone comparison to isolate cross-attention, and box-dropout to test the shortcut hypothesis are the first priorities. Validation on CBIS-DDSM and INbreast with RSNA fine-tuning (ii), and a prospective triage study with measured reader sensitivity, would then test generalisation and the assumed 85% reader.

V. CONCLUSION

MamNet-6 raises VinDr-Mammo AUROC from 0.7564 (best single-view CNN) to 0.8458 (16-model ensemble), an observed, untested 11.8% gain. Bounding-box outlines lowered AUROC by 0.12–0.20, plausibly through shortcut learning, and among same-backbone configurations CrossView had lower observed AUROC than BreastPairNet. MC Dropout entropy enables triage: 50% AI reads keep 87.9% sensitivity with an oracle reader but 77.3% with an 85%-sensitive reader, so K$\leq$40% is needed to stay above the 80% floor. RSNA 2022 (AUROC=0.6008) shows an open generalisation gap.

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